MetaOpt-AI | Antenna Design Explorer

Explainable Machine Learning for Metasurface Antenna Optimization

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WARNING - SYNTHETIC DATA. SYNTHETIC DATA: this dataset was generated from a documented analytical formula with added noise. It demonstrates the software workflow only. Results are NOT electromagnetic simulations or laboratory measurements and say nothing about real-world antenna accuracy.

1. Dataset and provenance

Field	Value
Dataset ID	1
Name	Synthetic demonstration dataset v1 (L_mm, G_mm)
Type	SYNTHETIC - Synthetic demonstration data (analytical formula + noise, NOT measured)
Samples	500
Inputs	L_mm, G_mm
Targets	S11_dB, Gain_dBi
SHA-256 checksum	cd5d5ec0f182d2ff9537299e41847020a7bfc9feb3751830448cefef26792b96
Registered	2026-10-19 16:27:33

Provenance: SYNTHETIC. Generated by datasets/synthetic.py (MetaOpt-AI) from a documented, physics-inspired analytical formula (half-wave resonance with assumed eps_eff=2.0, F0=10 GHz, Q=5, gap-dependent matching and metasurface gain term) plus seeded Gaussian noise. NOT an EM simulation, NOT a measurement.

Description: 500 Latin-hypercube samples, seed=42, L_mm in (8.0, 12.5), G_mm in (0.2, 2.0); noise sigma S11=0.6 dB, gain=0.15 dB. Demonstrates the workflow only.

variable	count	mean	std	min	max
L_mm	500	10.25	1.3	8.005	12.5
G_mm	500	1.1	0.5202	0.2006	1.999
S11_dB	500	-4.458	4.329	-24.56	-0.05
Gain_dBi	500	3.997	2.52	-2.963	8.273

2. Methodology

Data are validated (types, missing values, duplicates, IQR outliers flagged but not removed). Missing target values are never imputed; rows are excluded per target.

For each target a reproducible train/test split is made first. Scaling sits inside a scikit-learn Pipeline. K-fold cross-validation and RandomizedSearchCV tuning use only the training portion; the test set is used once for final metrics. The active model per target is chosen by CV R2, not test R2.

Models: Dummy (mean) baseline, Random Forest, Extra Trees, Gradient Boosting, Histogram Gradient Boosting and Gaussian Process regression (ARD kernel + white noise).

Optimization calls the trained surrogates inside SciPy Differential Evolution, LHS multi-start Nelder-Mead and Dual Annealing. Constraints use a static penalty on range-normalized violations; the trade-off objective adds range-normalized desirabilities so dB and dBi are never summed directly.

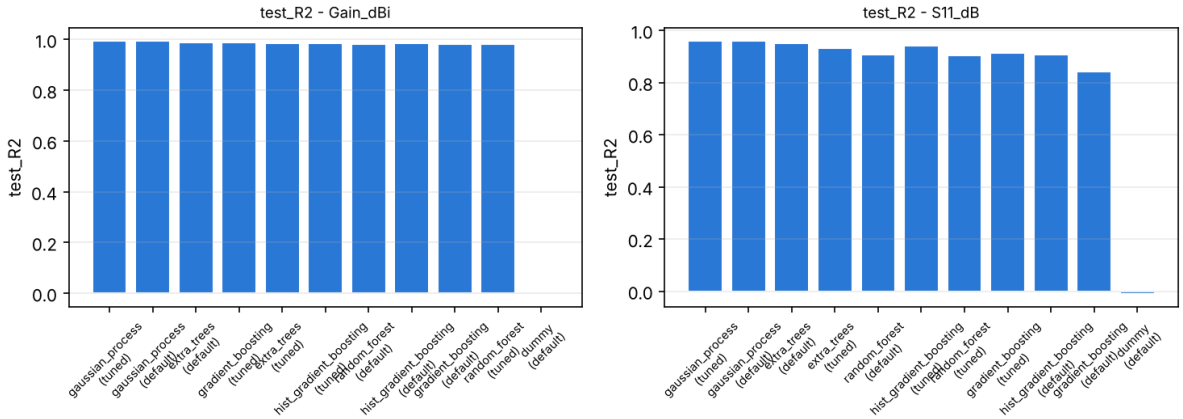
Explainability: impurity and permutation importance, one-at-a-time sensitivity, partial dependence. Uncertainty: GP predictive std or tree-ensemble spread, with empirical 95% interval coverage on the test set.

3. Model configurations and evaluation metrics

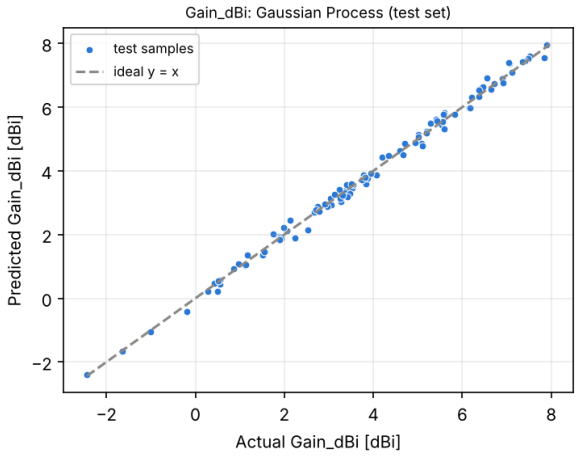
id	target	model	variant	test_R2	test_MAE	test_RMSE	cv_R2_mean	cv_R2_std	coverage95	n_train	n_test
22	Gain_dBi	gaussian_process	tuned	0.9954	0.1235	0.1562	0.9957	0.00113	0.94	400	100
21	Gain_dBi	gaussian_process	default	0.9954	0.1235	0.1562	0.9957	0.00113	0.94	400	100
15	Gain_dBi	extra_trees	default	0.9901	0.1818	0.2291	0.9888	0.002979	1	400	100
18	Gain_dBi	gradient_boosting	tuned	0.9882	0.194	0.25	0.9876	0.002461	-	400	100
16	Gain_dBi	extra_trees	tuned	0.9864	0.2092	0.2683	0.9842	0.005265	0.98	400	100
20	Gain_dBi	hist_gradient_boosting	tuned	0.9867	0.2076	0.2656	0.9829	0.005592	-	400	100
13	Gain_dBi	random_forest	default	0.9827	0.2321	0.3029	0.9796	0.006828	0.95	400	100
19	Gain_dBi	hist_gradient_boosting	default	0.984	0.2333	0.2915	0.978	0.007676	-	400	100
17	Gain_dBi	gradient_boosting	default	0.9826	0.2318	0.304	0.9766	0.007832	-	400	100
14	Gain_dBi	random_forest	tuned	0.981	0.2414	0.3174	0.9761	0.007579	0.94	400	100
12	Gain_dBi	dummy	default	-0.00427	1.924	2.309	-0.01488	0.01463	-	400	100
11	S11_dB	gaussian_process	tuned	0.9606	0.656	0.7863	0.9666	0.007151	0.95	400	100
10	S11_dB	gaussian_process	default	0.9592	0.6596	0.8004	0.9644	0.008612	0.96	400	100
4	S11_dB	extra_trees	default	0.951	0.7318	0.8772	0.9475	0.01959	0.84	400	100
5	S11_dB	extra_trees	tuned	0.931	0.7998	1.041	0.9237	0.02447	0.75	400	100
2	S11_dB	random_forest	default	0.9069	0.8796	1.208	0.9133	0.02503	0.83	400	100
9	S11_dB	hist_gradient_boosting	tuned	0.94	0.7521	0.9705	0.9054	0.02893	-	400	100
3	S11_dB	random_forest	tuned	0.9031	0.8993	1.233	0.9039	0.02842	0.8	400	100
7	S11_dB	gradient_boosting	tuned	0.9144	0.8768	1.159	0.8951	0.02117	-	400	100
8	S11_dB	hist_gradient_boosting	default	0.9068	0.8555	1.209	0.893	0.04217	-	400	100
6	S11_dB	gradient_boosting	default	0.8437	1.052	1.566	0.8457	0.04086	-	400	100
1	S11_dB	dummy	default	-0.01236	3.167	3.985	-0.006041	0.005243	-	400	100

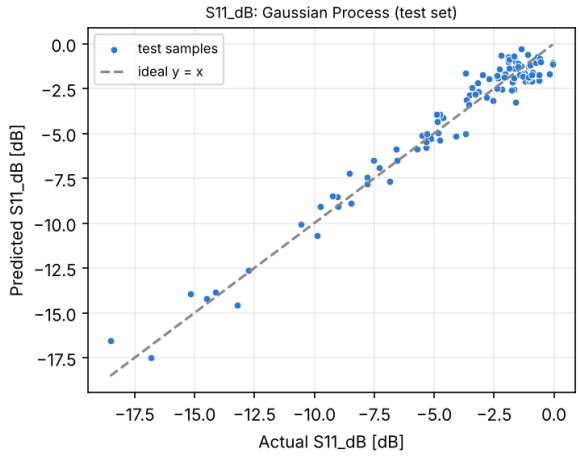
MAE/RMSE are in the target unit (dB for S11, dBi for gain). coverage95 = fraction of test points inside +/-1.96 sigma (models with uncertainty only).

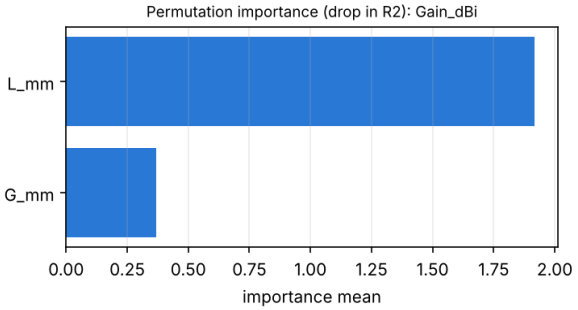
Active models (selected by cross-validated R2 on training data): Gain_dBi: #22 Gaussian Process (tuned), S11_dB: #11 Gaussian Process (tuned)

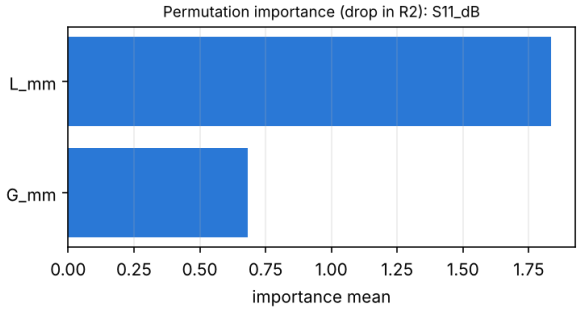


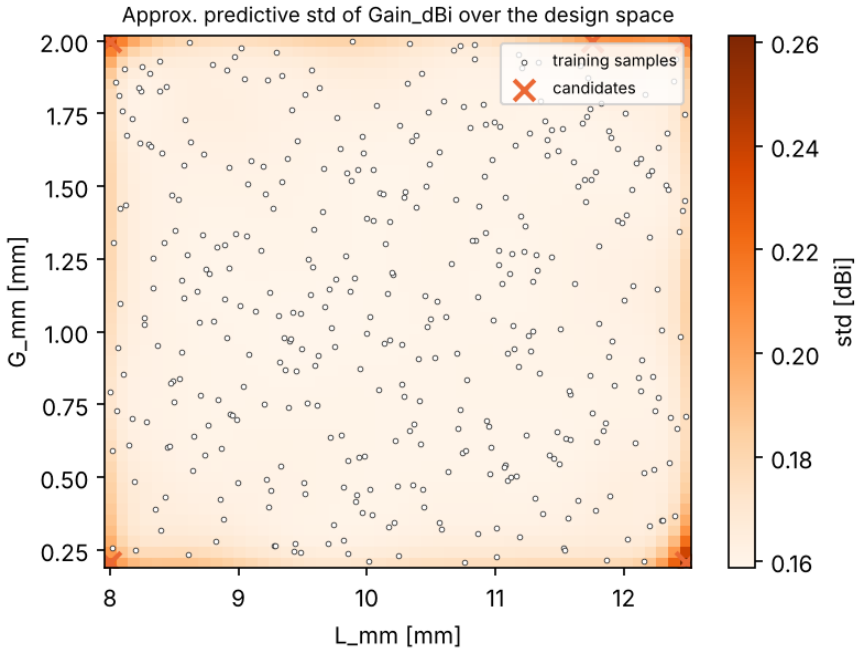
4. Figures for the active models









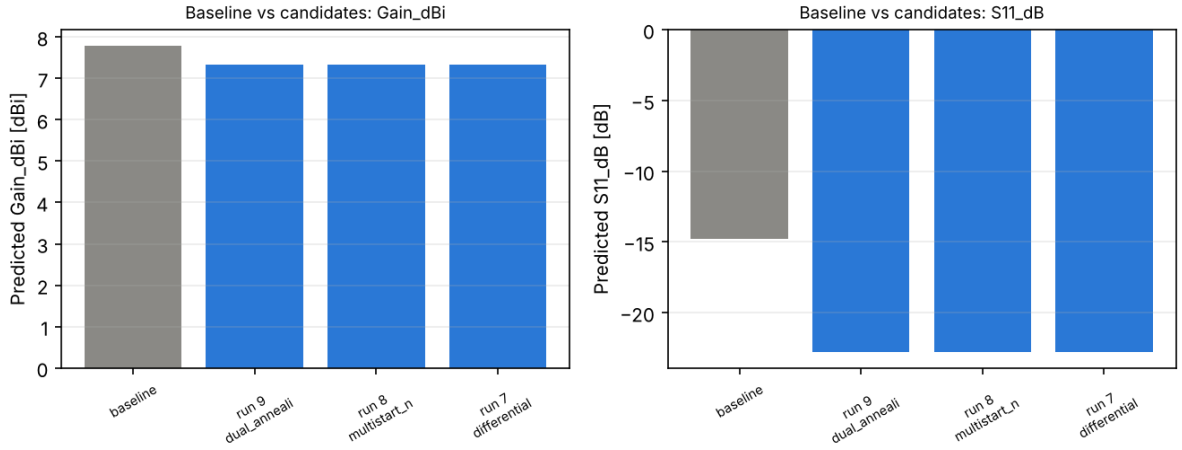


Crosses mark the highest-uncertainty regions: candidate locations for additional EM simulations or measurements. This is a suggestion; no improvement in accuracy from such sampling has been demonstrated in this project. Uncertainty values are APPROXIMATE model-based indicators (GP predictive std or tree-ensemble spread). They are not calibrated confidence intervals and do not include errors of the synthetic data-generating formula or of real EM physics.

5. Optimization results (ML-selected candidates)

run	algorithm	objective	constraints	candidate	predicted	feasible	evals
9	dual_annealing	tradeoff	S11_dB <= -10.0	L_mm=9.948, G_mm=0.877	Gain_dBi=7.34, S11_dB=-22.76	True	1500
8	multistart_nelder_mead	tradeoff	S11_dB <= -10.0	L_mm=9.945, G_mm=0.876	Gain_dBi=7.33, S11_dB=-22.77	True	901
7	differential_evolution	tradeoff	S11_dB <= -10.0	L_mm=9.945, G_mm=0.876	Gain_dBi=7.33, S11_dB=-22.77	True	990
6	dual_annealing	min_s11	Gain_dBi >= 7.0	L_mm=9.935, G_mm=0.856	Gain_dBi=7.29, S11_dB=-22.81	True	1500
5	multistart_nelder_mead	min_s11	Gain_dBi >= 7.0	L_mm=9.936, G_mm=0.853	Gain_dBi=7.29, S11_dB=-22.82	True	899
4	differential_evolution	min_s11	Gain_dBi >= 7.0	L_mm=9.936, G_mm=0.853	Gain_dBi=7.29, S11_dB=-22.82	True	1050

Most recent run #9: baseline {'L_mm': 10.2536, 'G_mm': 1.10025} -> predicted {Gain_dBi: 7.784, S11_dB: -14.803}; candidate {L_mm: 9.948, G_mm: 0.877} -> predicted {Gain_dBi: 7.337, S11_dB: -22.755} (approx. std {Gain_dBi: 0.160, S11_dB: 0.694}). Bounds: {'L_mm': [8.0052, 12.482], 'G_mm': [0.2064, 1.9992]}.



All candidates are ML-selected designs based on surrogate predictions. They are NOT physically validated antenna designs.

6. Limitations

• The surrogate models are statistical regressions; they do not solve Maxwell's equations and are not a substitute for full-wave EM simulation or measurement.

• When the dataset is SYNTHETIC, all metrics measure how well the models recover a documented analytical formula plus noise. They say nothing about real antenna accuracy.

• Optimized geometries are ML-selected candidates. They must be verified by EM simulation and prototype measurement before any engineering claim is made.

• A deeper S11 alone does not prove better gain, bandwidth, efficiency or radiation pattern.

• Uncertainty estimates (GP std, tree spread) are approximate and only empirically checked on the held-out test split; they are not calibrated confidence intervals.

• Predictions outside the training range are extrapolations and are flagged as unreliable.

• Only two geometry parameters and two targets are modelled in the demonstration; real designs depend on substrate, feed, unit-cell shape, frequency and fabrication tolerances.

7. Reproducibility information

Item	Value
Dataset checksum (SHA-256)	cd5d5ec0f182d2ff9537299e41847020a7bfc9feb3751830448cefef26792b96
Global random seed	42
Python	3.13.16

Item	Value
scikit-learn	1.9.1
SciPy	1.18.1
NumPy	2.5.3
pandas	3.0.5
Platform	Linux-6.18.44-fc-v139-x86_64-with-glibc2.39

Gain_dBi - experiment #22 gaussian_process (tuned), seed 42, validation KFold(n_splits=5, shuffle=True), hyperparameters: alpha=1e-10, copy_X_train=True, kernel=1**2 * RBF(length_scale=[1, 1]) + WhiteKernel(noise_level=0.01), n_restarts_optimizer=2, n_targets=None, normalize_y=True, optimizer=fmin_l_bfgs_b, random_state=42

S11_dB - experiment #11 gaussian_process (tuned), seed 42, validation KFold(n_splits=5, shuffle=True), hyperparameters: alpha=1e-10, copy_X_train=True, kernel=1**2 * Matern(length_scale=[1, 1], nu=1.5) + WhiteKernel(noise_level=0.01), n_restarts_optimizer=2, n_targets=None, normalize_y=True, optimizer=fmin_l_bfgs_b, random_state=42

Optimization #9 dual_annealing, seed 42, budget 1500 (used 1500), settings maxfun=1500, no_local_search=True, visit=2.62, accept=-5.0

Optimization #8 multistart_nelder_mead, seed 42, budget 1500 (used 901), settings n_screen_lhs=600, n_starts=5, maxfev_per_start=180, xatol=0.0001, fatol=1e-06

Optimization #7 differential_evolution, seed 42, budget 1500 (used 990), settings popsize_multiplier=15, population=30, maxiter=49, mutation=[0.5, 1.0], recombination=0.7, tol=1e-06, polish=False, init=latinhypercube, strategy=best1bin